Database Schema — DDI Model (DDI 4 / DDI-CDI / DDI-L 3.3)

> **Status:** Implementation Complete & Validated
> **Scope:** Standard-agnostic core DDI data model for the ReQuest question bank, aligned with DDI 4 (COGS model), DDI-CDI, and DDI-Lifecycle 3.3 with canonical URN-based primary keys
> **Target DB:** PostgreSQL ≥ 17 (ICU collations, JSONB indexing, standard SQL DDL; compatible with SQLite dev & test)
> **Interactive Model Explorer:** `database_explorer.html` *(interactive browser app)*
> **Full Mermaid Diagrams:** `database_diagram.md` | `database_diagram.mmd`

---

1. Architecture & Design Conventions

The FAIRwDDI architecture implements a **standard-agnostic, lightweight DDI model** optimized for social-science question banks and cross-survey harmonization. Instead of monolithic specification bloat, the data model organizes metadata into clean, decoupled layers:

1. **Organization & Study Grouping Layer:** Institutional agencies, archives, and polymorphic resource groupings (study series, panels, thematic collections).
2. **Concept Layer:** Controlled vocabularies (SKOS/ELSST), concept hierarchies, and abstract conceptual variables.
3. **Representation & Instrument Layer:** Decoupled question wording, reusable categories, coded response domains, represented variables, and data collection instruments.
4. **Dataset Layer:** Physical survey waves (`StudyUnit`) and concrete data columns (`InstanceVariable`).
5. **Cross-Layer Dynamics:** The DDI Variable Cascade (`ConceptualVariable → RepresentedVariable → InstanceVariable`) and semantic graph relationships.
6. **Infrastructure, Staging & Audit Layer:** Streaming ingestion staging, URN aliasing, metadata quarantine, and lifecycle event logging.

1.1 Conventions Summary

Convention	Rule
Supertype Master Registry	`request_ddi_urnregistry` — master table of all canonical URNs and external URIs (`urn`, `resource_type`, `extended_attributes`).
Primary key (DDI Resources)	`urn` — `VARCHAR(512) PRIMARY KEY REFERENCES request_ddi_urnregistry(urn) ON DELETE CASCADE` (Canonical DDI 3.3 / 4.0 URN format).
Primary key (Junction & Non-DDI)	`id` — `BigAutoField` / `BIGSERIAL PRIMARY KEY`.
Parent Scheme Foreign Keys	`scheme_urn` — `VARCHAR(512) REFERENCES ...(urn)` (maps to `models.ForeignKey(..., db_column="scheme_urn")`).
Inter-Resource Foreign Keys	`*_urn` — `VARCHAR(512) REFERENCES ...(urn)` (e.g. `study_unit_urn`, `represented_variable_urn`, `conceptual_variable_urn`, `question_item_urn`, `code_list_urn`, `category_urn`, `instrument_urn`).
Polymorphic Foreign Keys	`VARCHAR(512) REFERENCES request_ddi_urnregistry(urn) ON DELETE CASCADE` (used in `URNAlias.canonical_urn`, `SemanticRelationship.subject_urn/object_urn`, `EventLog.urn`).
Canonical URN Format	Colon-separated: `urn:ddi:agency[.sub-agency]:ID:Version`. • Agency-scoped: `urn:ddi:fr.sciencespo:V321:1.0.0` • Maintainable-scoped: `urn:ddi:fr.sciencespo:MaintainableID.ObjectID:1.0.0`
Unified Hashes	`hashes` (`JSONB`) — unified key-value mapping of hash algorithm types to digests (e.g. `{"sha256": "...", "canonical_nfkc": "..."}`).
Extended Attributes	`extended_attributes` (`JSONB`) — array of objects capturing flexible provider/specification attributes (intent, interviewer guidance, notes, value domains, inclusion/exclusion) without SQL schema bloat.
Multilingual Text	`JSONB` array of objects: `[{"lang": "fr", "value": "...", "type": "literal"}]`. Extensible attributes supported.
Timestamps	`created_at` (`auto_now_add`), `updated_at` (`auto_now`) on all entity tables.
Naming	Table names use canonical DDI model terminology (`request_ddi_{snake_case}`).

1.2 DDI Resource Base Models & Supertype Lifecycle

Every DDI entity inherits from `DDIIdentifiable` providing persistent canonical URN identification and automatic supertype synchronization with `UrnRegistry`. Scheme container packages inherit from `DDIScheme` (`urn`, `name`, `description`, `extended_attributes` without hashes), while versionable DDI resources inherit from `DDIResource`, where `urn` and `name` are required, multi-algorithm content `hashes` are maintained, and all entity-specific metadata fields are nullable:

```
class DDIIdentifiable(models.Model):
    """Abstract mixin for DDI-Lifecycle URN identification and UrnRegistry supertype sync."""

    urn = models.CharField(
        max_length=512,
        primary_key=True,
```

```
        help_text="Persistent canonical URN: urn:ddi:{agency}:{ID}:{version}",
    )
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)

    class Meta:
        abstract = True

    def save(self, *args, **kwargs) -> None:
        """Ensure canonical URN is generated and pre-registered in UrnRegistry supertype table."""
        if not self.urn:
            agency = os.getenv("DDI_AGENCY", "fr.sciencespo")
            primary_hash = None
            hashes_val = getattr(self, "hashes", None)
            if isinstance(hashes_val, dict):
                primary_hash = hashes_val.get("sha256") or hashes_val.get("primary")
            object_id = (
                primary_hash[:16]
                if primary_hash
                else f"{self.__class__.__name__}-{uuid.uuid4().hex[:12]}"
            )
            self.urn = f"urn:ddi:{agency}:{object_id}:1.0.0"

        # Pre-register URN in UrnRegistry supertype table before inserting child row
        from fairwddi.models.infrastructure import UrnRegistry

        UrnRegistry.objects.update_or_create(
            urn=self.urn,
            defaults={"resource_type": self.__class__.__name__},
        )
        super().save(*args, **kwargs)

    def delete(self, *args, **kwargs):
        """Delete instance and clean up central UrnRegistry record."""
        from fairwddi.models.infrastructure import UrnRegistry

        urn = self.urn
        res = super().delete(*args, **kwargs)
        UrnRegistry.objects.filter(urn=urn).delete()
        return res
```

class DDIScheme(DDIIdentifiable):
"""Abstract base model for all DDI scheme containers (no content hashes)."""

```
name = models.JSONField(
    default=list,
    help_text="Multilingual scheme name: [{'lang': 'fr', 'value': '...' }].",
)
description = models.JSONField(
    default=list,
    blank=True,
    null=True,
    help_text="Multilingual description.",
)
extended_attributes = models.JSONField(
    default=list,
    blank=True,
    help_text="Extensible attributes list of objects.",
```

```
)

class Meta:
    abstract = True
```

class DDIResource(DDIIdentifiable):
"""Abstract base model for all top-level DDI resources requiring urn and name, with hashes."""

```
name = models.JSONField(
    default=list,
    help_text="Multilingual name: [{'lang': 'fr', 'value': '...' }].",
)
hashes = models.JSONField(
    default=dict,
    blank=True,
    null=True,
    help_text="Unified key-value hash digests dictionary (e.g. {'sha256': '...'}).",
)

class Meta:
    abstract = True
```

```
---

## 2. Organization & Study Grouping Layer

The Organization & Grouping layer manages institutional actors (distributors, archives, funding
bodies) and logical resource aggregations (longitudinal survey series, panels, thematic collections)
without rigid physical foreign-key coupling.

```mermaid
graph TD
    Org["<b>Organization</b><br/><i>(Archive, Funder, Distributor)</i>"]
    Grp["<b>Group</b><br/><i>(study_series, panel, thematic)</i>"]
    SU["<b>StudyUnit</b><br/><i>(Survey wave / dataset)</i>"]
    Grp -.->|polymorphic references| SU
```

2.1 Table Definitions

Organization

Institutional organization resource in DDI-Lifecycle (maps to DDI-L `<Organization>`). Represents archives, distributors, research centers, universities, and funding bodies. Secondary details (acronyms, contact info, URIs) are captured in `extended_attributes`. Inherits `DDIResource`.
```

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical DDI-L Organization URN
`name`	`JSONField`	required	Multilingual organization name `[{"lang": "fr", "value": "..."}]`
`organization_type`	`CharField(64)`	default `'distributor'`, indexed	Classification type (`'archive'`, `'distributor'`, `'funder'`, etc.)
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`hashes`	`JSONField`	default `{}`	Multi-algorithm content digests
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

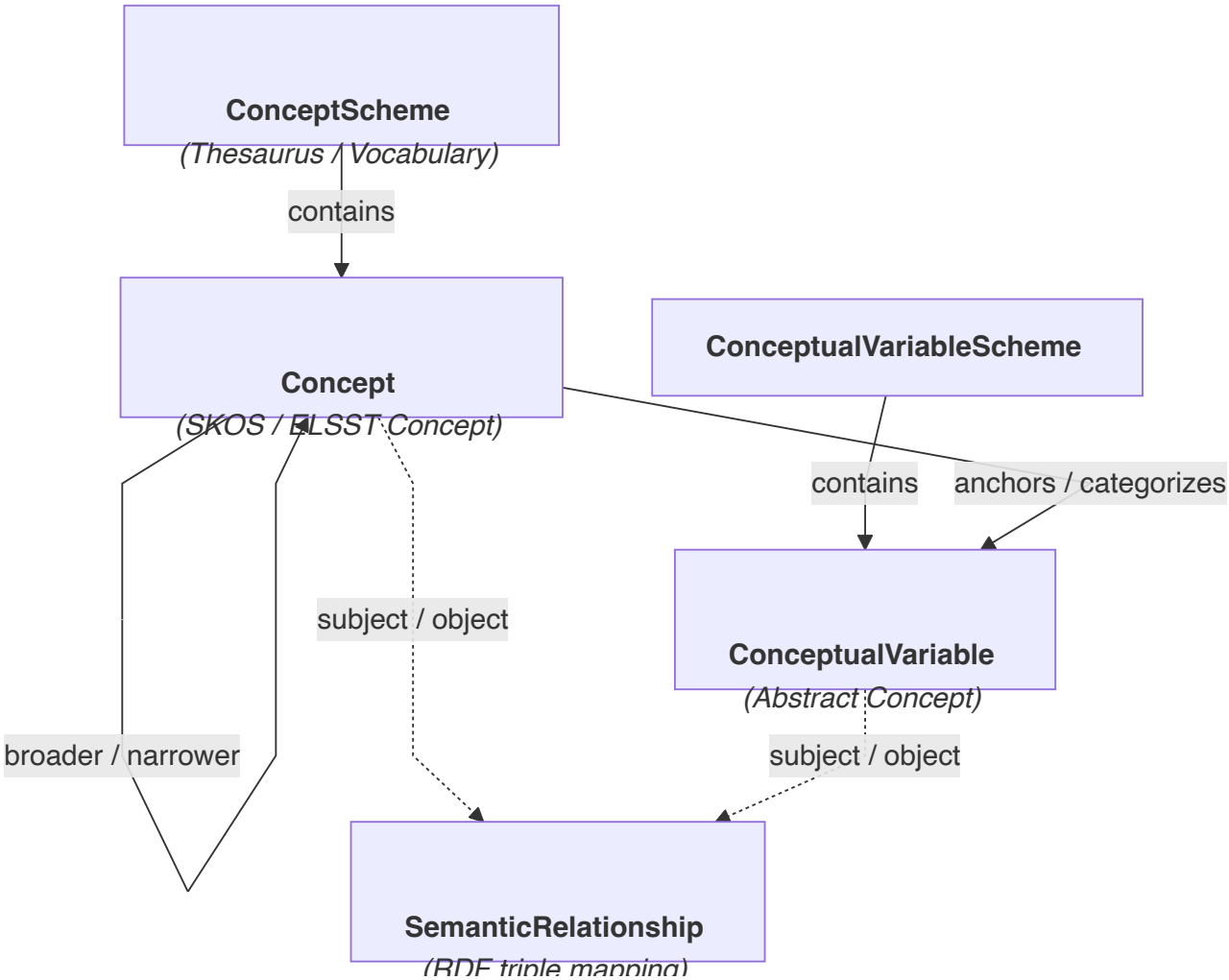
Group

Generic Group DDI resource (maps to DDI-L `<Group>` / `<SubGroup>`). Logical grouping of related resources (e.g. study series, panels, thematic collections, or longitudinal waves) referenced via polymorphic typed URN references in the `references` JSON field. Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical DDI-L Group URN
`name`	`JSONField`	required	Multilingual group name `[{"lang": "fr", "value": "..."}]`
`description`	`JSONField`	nullable	Multilingual group description
`group_type`	`CharField(64)`	default `'study_series'`, indexed	Group classification type (`'study_series'`, `'panel'`, `'thematic'`, `'collection'`)
`references`	`JSONField`	default `[]`	Member resource references `[{"resource_type": "StudyUnit", "urn": "..."}]`
`hashes`	`JSONField`	default `{}`	Multi-algorithm content digests
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

3. Concept Layer

The Concept layer anchors survey metadata to formal thesauri (CESSDA ELSST, CESSDA Topics, DDI-CV) and establishes high-level measurement concepts independently of question wording or dataset implementations.



3.1 Table Definitions

ConceptScheme

Named collection of reusable concepts (maps to DDI-L `ConceptScheme` / SKOS `skos:ConceptScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical ConceptScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

Concept

High-level thematic domain concept from any controlled vocabulary or thesaurus (e.g. CESSDA ELSST, CESSDA Topics, DDI-CV). Inherits `DDIIdentifiable`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical Concept URN
`scheme_urn`	`CharField(512)`	FK → ConceptScheme(urn), nullable	Parent ConceptScheme defining this concept
`uri`	`CharField(512)`	unique, nullable	Controlled vocabulary concept URI (e.g. ELSST URI)
`vocabulary`	`CharField(128)`	default `""`	Controlled vocabulary name (e.g. `"ELSST"`, `"Topics"`)
`notation`	`CharField(128)`	nullable	Standard thesaurus notation code
`label`	`JSONField`	nullable	Multilingual preferred label (`skos:prefLabel`)
`description`	`JSONField`	default `[]`	Multilingual description
`definition`	`JSONField`	default `[]`	Multilingual definition (`skos:definition`)
`parent_urn`	`CharField(512)`	FK → Concept(urn), nullable	Parent concept for hierarchical trees (`skos:broader`)
`concept_type`	`CharField(64)`	default `'concept'`	Classification type (`'domain'`, `'concept'`, `'top_concept'`)
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

ConceptualVariableScheme

Named collection of reusable conceptual variables (maps to DDI-L `ConceptualVariableScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical ConceptualVariableScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

ConceptualVariable

Abstract measurement concept (e.g., "Left-Right Political Placement", "Subjective Health"). Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical URN (`urn:ddi:agency:cv-...:1.0.0`)
`scheme_urn`	`CharField(512)`	FK → ConceptualVariableScheme(urn), nullable	Parent ConceptualVariableScheme
`concept_urn`	`CharField(512)`	FK → Concept(urn), nullable	Controlled vocabulary concept anchor
`name`	`JSONField`	required	Multilingual / faceted conceptual variable name
`label`	`JSONField`	nullable	Multilingual human label
`description`	`JSONField`	nullable	Multilingual description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

SemanticRelationship

Captures RDF triple-like relationships between resources across any layer (e.g. SKOS mappings `skos:exactMatch`, `skos:broadMatch`, `skos:relatedMatch`, lineage links `prov:wasDerivedFrom`, cross-vocabulary thesaurus anchors).

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing identifier
`subject_type`	`CharField(128)`	indexed	Subject entity type (`'Concept'`, `'ConceptualVariable'`, etc.)
`subject_urn`	`CharField(512)`	indexed	Canonical URN or URI of the subject resource
`predicate`	`CharField(256)`	indexed	Semantic predicate (e.g. `'skos:exactMatch'`, `'skos:broadMatch'`)
`object_type`	`CharField(128)`	indexed	Target entity type (`'Concept'`, `'ConceptualVariable'`, etc.)
`object_urn`	`CharField(512)`	indexed	Canonical URN or URI of the target resource
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects (confidence, tool, notes)
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

Constraints & Indexes:

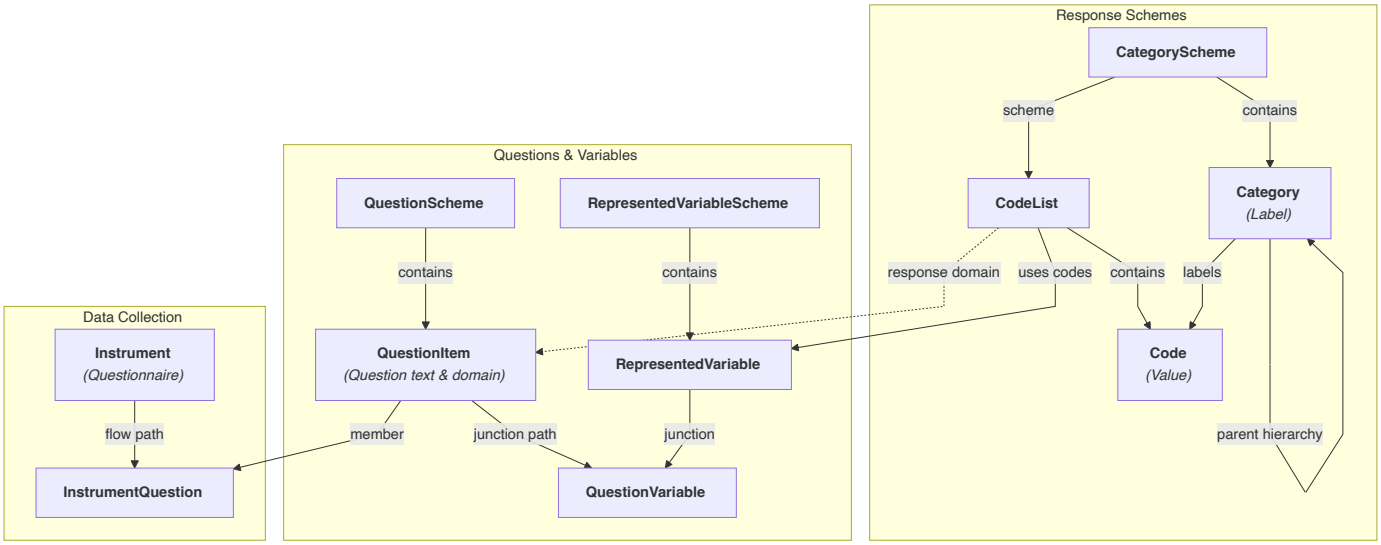
- Unique constraint: `(subject_urn, predicate, object_urn)`

- Indexes: `(subject_urn, predicate)`, `(object_urn, predicate)`, `(predicate)`, `(subject_type)`, `(object_type)`

---

4. Representation & Instrument Layer

The Representation & Instrument layer defines reusable response schemes, standalone question texts, harmonized represented variables, and data collection questionnaires.



4.1 Response Representation: Categories & Code Lists

CategoryScheme

Named collection of reusable categories (maps to DDI-L `CategoryScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical CategoryScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

Category

Response category text label (decoupled from numerical code values). Belongs to a single `CategoryScheme` and supports self-referential hierarchy. Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical Category URN
`scheme_urn`	`CharField(512)`	FK → CategoryScheme(urn), nullable	Parent CategoryScheme defining this category
`concept_urn`	`CharField(512)`	FK → Concept(urn), nullable	Optional semantic concept anchor in a controlled vocabulary
`name`	`JSONField`	required	Technical identifier / multilingual name
`label`	`JSONField`	nullable	Multilingual category label
`parent_urn`	`CharField(512)`	FK → Category(urn), nullable	Parent category for hierarchical schemes
`order`	`PositiveIntegerField`	default 0	Display order within scheme
`is_missing`	`BooleanField`	default `False`	Missing / non-response indicator
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Category definitions, inclusions/exclusions
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

CodeList

Structural set of response codes linked to categories. Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical CodeList URN
`scheme_urn`	`CharField(512)`	FK → CategoryScheme(urn), nullable	Optional link to parent CategoryScheme
`name`	`JSONField`	required	Multilingual human title
`description`	`JSONField`	nullable	Multilingual description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

Code

Junction connecting a `CodeList` to a `Category` with a discrete code value. Inherits `DDIIdentifiable`. Supports self-referential hierarchy (`parent_urn`).

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Persistent canonical URN: `urn:ddi:{agency}:{ID}:{version}`
`code_list_urn`	`CharField(512)`	FK → CodeList(urn)	Parent code list
`category_urn`	`CharField(512)`	FK → Category(urn)	Referenced category label
`code_value`	`CharField(64)`	required	Numerical/string code (e.g. `"1"`, `"98"`)
`parent_urn`	`CharField(512)`	FK → Code(urn), nullable	Parent code for hierarchical code lists
`order`	`PositiveIntegerField`	default 0	Display order within list
`is_missing`	`BooleanField`	default `False`	True if code represents missing/non-response (DK, Refusal, NA)
`hashes`	`JSONField`	default `{}`	Multi-algorithm content digests
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

4.2 Question Representation

QuestionScheme

Named collection of reusable questions (maps to DDI-L `QuestionScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical QuestionScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

QuestionItem

Standalone reusable question wording and response domain specification. Belongs to a single `QuestionScheme`, can link directly to a response `CodeList` or define a flexible `response_domain` JSON object (coded, numeric, text, datetime, scale, etc.), and inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical QuestionItem URN
`scheme_urn`	`CharField(512)`	FK → QuestionScheme(urn), nullable	Parent QuestionScheme defining this question
`code_list_urn`	`CharField(512)`	FK → CodeList(urn), nullable	Associated response CodeList (null for non-coded response domains)
`response_domain`	`JSONField`	default `{}`	JSON object describing response domain (`{"type": "code", "code_list_urn": "..."}`, `{"type": "numeric", ...}`)
`name`	`JSONField`	required	Technical identifier / multilingual name
`question_text`	`JSONField`	nullable	Multilingual literal question text
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Preamble, interviewer instructions, intent
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

4.3 Represented Variables & Question Binding

RepresentedVariableScheme

Named collection of reusable represented variables (maps to DDI-L `RepresentedVariableScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical RepresentedVariableScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

RepresentedVariable

Combination of conceptual variable and response representation (e.g., code list, numeric range, text domain). Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical RepresentedVariable URN
`scheme_urn`	`CharField(512)`	FK → RepresentedVariableScheme(urn), nullable	Parent RepresentedVariableScheme
`conceptual_variable_urn`	`CharField(512)`	FK → ConceptualVariable(urn), nullable	Parent concept
`code_list_urn`	`CharField(512)`	FK → CodeList(urn), nullable	Direct response code structure foreign key
`value_representation`	`JSONField`	default `{}`	JSON object describing value representation (`{"type": "code", "code_list_urn": "..."}`)
`name`	`JSONField`	required	Multilingual / faceted represented variable name
`label`	`JSONField`	nullable	Multilingual short label
`description`	`JSONField`	nullable	Multilingual description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Value domain, data type, variable characteristics
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

QuestionVariable

Junction connecting a `QuestionItem` to a `RepresentedVariable` with referencing path.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing junction ID
`question_item_urn`	`CharField(512)`	FK → QuestionItem(urn)	Associated question
`represented_variable_urn`	`CharField(512)`	FK → RepresentedVariable(urn)	Associated variable
`path`	`CharField(512)`	default `""`	Referencing path (e.g. XPath/DDI reference path)
`order`	`PositiveIntegerField`	default 0	Association order
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

4.4 Data Collection Instruments

Instrument

Data collection instrument / questionnaire in DDI-Lifecycle. Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical Instrument URN
`name`	`JSONField`	required	Multilingual technical name
`label`	`JSONField`	nullable	Multilingual human title
`description`	`JSONField`	nullable	Multilingual description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Mode of collection, administration notes
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

InstrumentQuestion

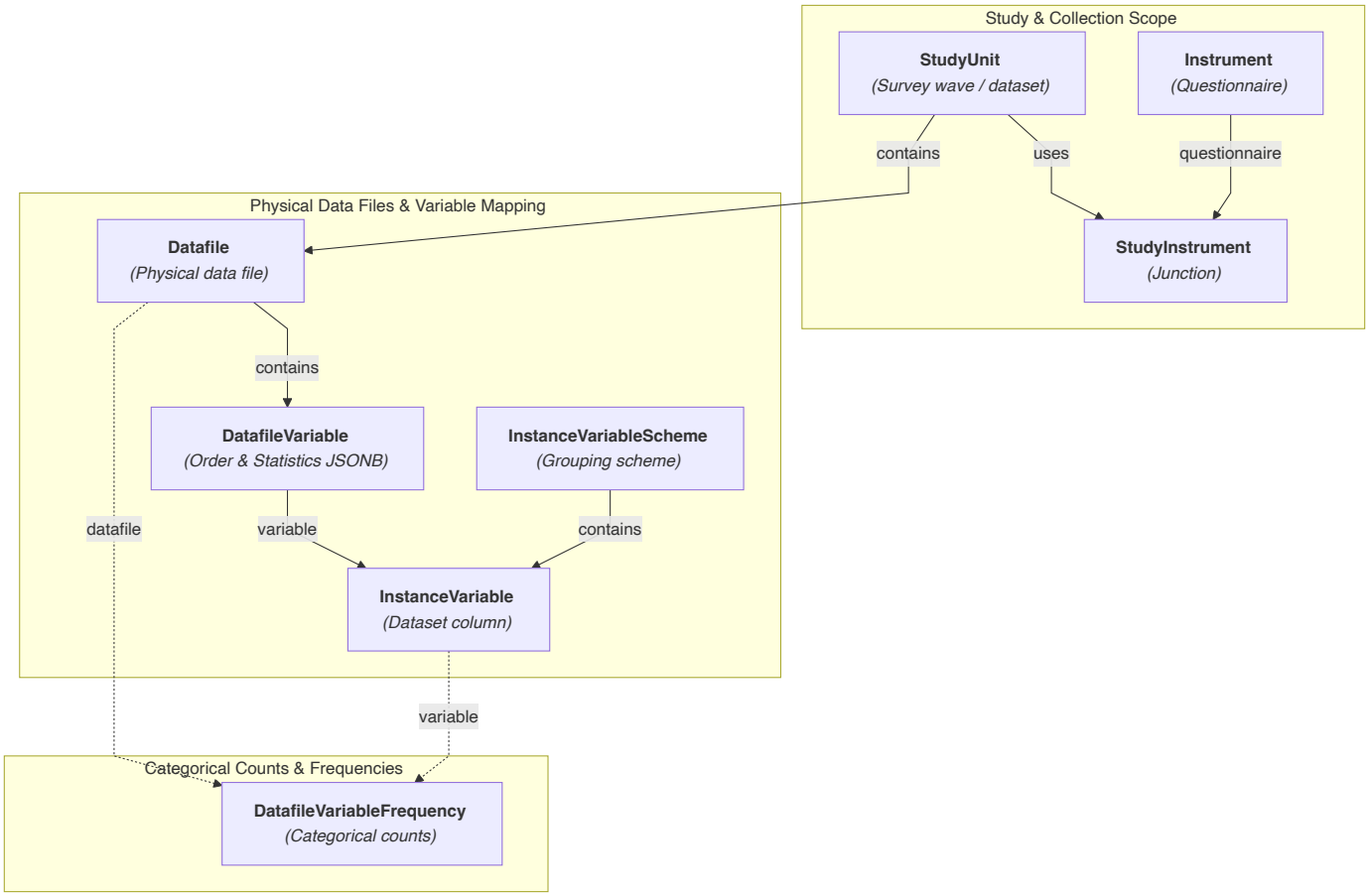
Junction connecting an `Instrument` to member `QuestionItem` entities with sequence/flow path.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing junction ID
`instrument_urn`	`CharField(512)`	FK → Instrument(urn)	Parent instrument
`question_item_urn`	`CharField(512)`	FK → QuestionItem(urn)	Associated question
`path`	`CharField(512)`	default `""`	Sequencing/flow path (e.g. `/Instrument/Sequence/Q01`)
`order`	`PositiveIntegerField`	default 0	Sequence display order
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

---

5. Dataset Layer

The Dataset layer represents specific survey data files, data collection instruments, and concrete physical column variables with categorical frequency distributions.



5.1 Table Definitions

StudyUnit

A specific survey wave or dataset (renamed from `Survey`). Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical StudyUnit URN
`name`	`JSONField`	required	Multilingual dataset identifier / name
`title`	`JSONField`	nullable	Multilingual study title
`external_ref`	`CharField(512)`	unique, nullable	DOI or external repository identifier
`year`	`PositiveIntegerField`	nullable	Survey reference year
`description`	`JSONField`	nullable	Multilingual abstract
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Methodological metadata
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

Datafile

A minimal Datafile entity relating `InstanceVariable` to `StudyUnit` via `DatafileVariable` (replaces legacy `StudyVariable`). A datafile points to a study wave. Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical Datafile URN
`study_unit_urn`	`CharField(512)`	FK → StudyUnit(urn)	Parent study wave
`name`	`JSONField`	required	File name / multilingual identifier
`description`	`JSONField`	nullable	Multilingual file description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	File format, case count, storage path
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

DatafileVariable

Junction table relating a `Datafile` to its member `InstanceVariable` columns, with display/column ordering and descriptive statistics JSONB.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing identifier
`datafile_urn`	`CharField(512)`	FK → Datafile(urn)	Parent Datafile
`instance_variable_urn`	`CharField(512)`	FK → InstanceVariable(urn)	Member InstanceVariable column
`order`	`IntegerField`	default 0	Column sequence / display order within datafile
`statistics`	`JSONField`	default `{}`	Descriptive summary statistics key-value pairs (mean, min, max, stddev, valid_cases)
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

InstanceVariableScheme

Named collection of instance variables (maps to DDI-L `InstanceVariableScheme` / `VariableScheme`). Inherits `DDIScheme`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical InstanceVariableScheme URN
`name`	`JSONField`	required	Multilingual scheme name
`description`	`JSONField`	nullable	Multilingual description
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

InstanceVariable

Physical realization of a variable in a specific dataset column within a `Datafile` (linked via `DatafileVariable`). Inherits `DDIResource`.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical InstanceVariable URN
`scheme_urn`	`CharField(512)`	FK → InstanceVariableScheme(urn), nullable	Parent InstanceVariableScheme
`represented_variable_urn`	`CharField(512)`	FK → RepresentedVariable(urn), nullable	Harmonized variable instantiated
`name`	`JSONField`	required	Multilingual / faceted dataset column name (e.g., `q01a`)
`label`	`JSONField`	nullable	Multilingual variable label
`description`	`JSONField`	nullable	Multilingual description
`hashes`	`JSONField`	default `{}`	Multi-algorithm digests
`extended_attributes`	`JSONField`	default `[]`	Universe, variable notes, storage format
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

DatafileVariableFrequency

Stores categorical variable count statistics per response code within a specific `Datafile`, with optional weighting.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing identifier
`datafile_urn`	`CharField(512)`	FK → Datafile(urn)	Datafile containing the variable observations
`instance_variable_urn`	`CharField(512)`	FK → InstanceVariable(urn)	Instance variable measured
`code_value`	`CharField(128)`	required	Discrete code value as string (e.g. `'1'`, `'98'`, `'DK'`)
`frequency`	`FloatField`	required	Count or frequency statistic
`weight`	`CharField(256)`	nullable	Optional weight variable name, URN, or specification
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

StudyInstrument

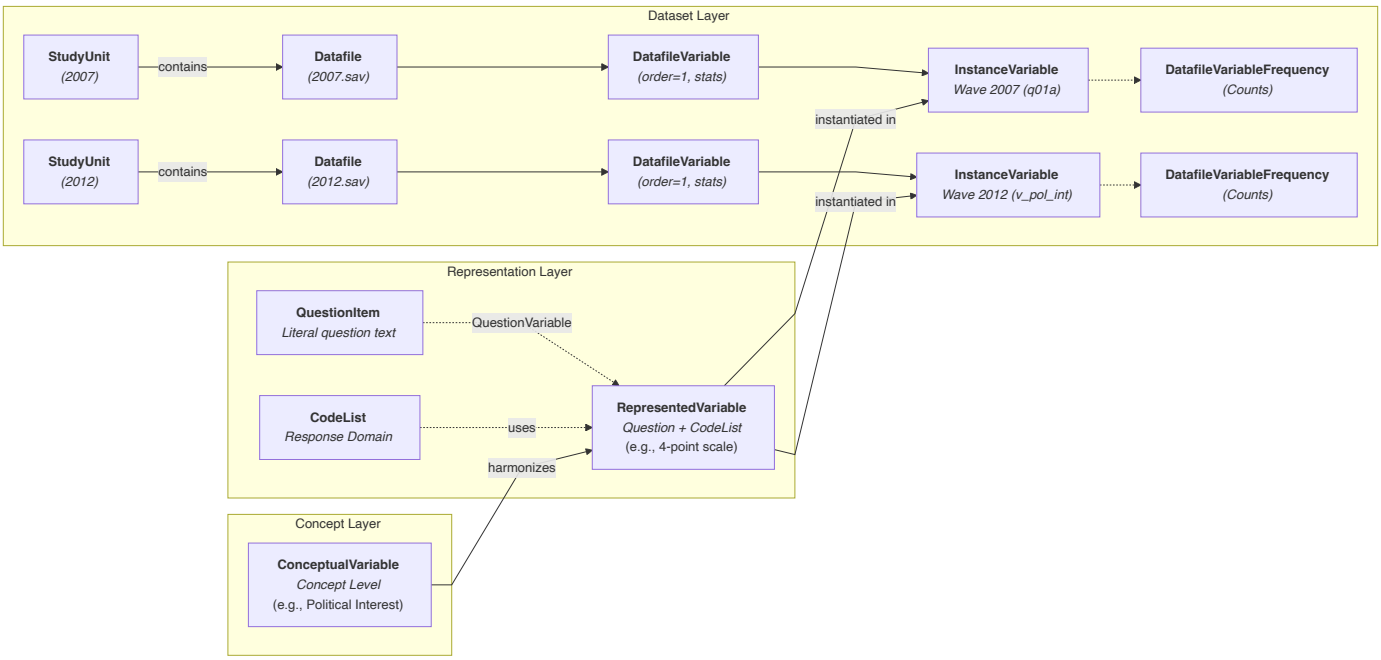
Junction capturing the direct relationship between a `StudyUnit` and its data collection `Instrument` with sequencing path.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing junction ID
`study_unit_urn`	`CharField(512)`	FK → StudyUnit(urn)	Study unit that used this instrument
`instrument_urn`	`CharField(512)`	FK → Instrument(urn)	Instrument used
`path`	`CharField(512)`	default `""`	Referencing path
`order`	`PositiveIntegerField`	default 0	Display order
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

---

6. Cross-Layer Dynamics & The Variable Cascade

The core innovation of the upgraded FAIRwDDI architecture is the **three-tier DDI Variable Cascade**, which decouples abstract concepts from measurement representations and physical dataset storage.

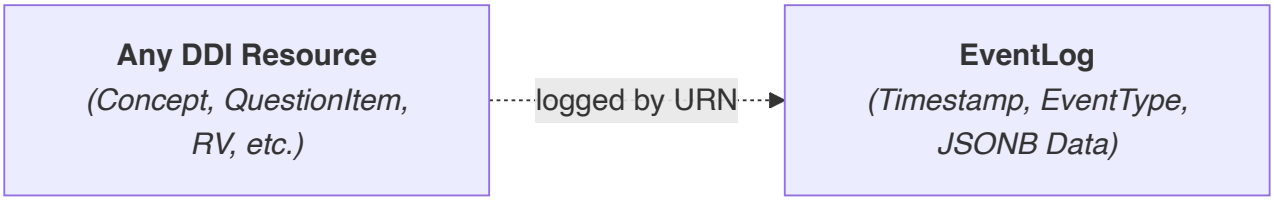


6.1 Harmonization Cascade Mechanics

1. **Top Tier — `ConceptualVariable`:** Defines *what* is measured (e.g., "Left-Right Political Scale") anchored to a controlled thesaurus `Concept`.
2. **Middle Tier — `RepresentedVariable`:** Pairs a question text (`QuestionItem`) with a standardized response code list (`CodeList`). Across multiple longitudinal waves, whenever the identical question text and response categories are used, they reference the **same** `RepresentedVariable`.
3. **Bottom Tier — `InstanceVariable`:** Represents the concrete physical column (e.g. `q01a` in 2007, `v_pol_int` in 2012) in a specific `StudyUnit`.

---

7. Event Logging & Audit Trail



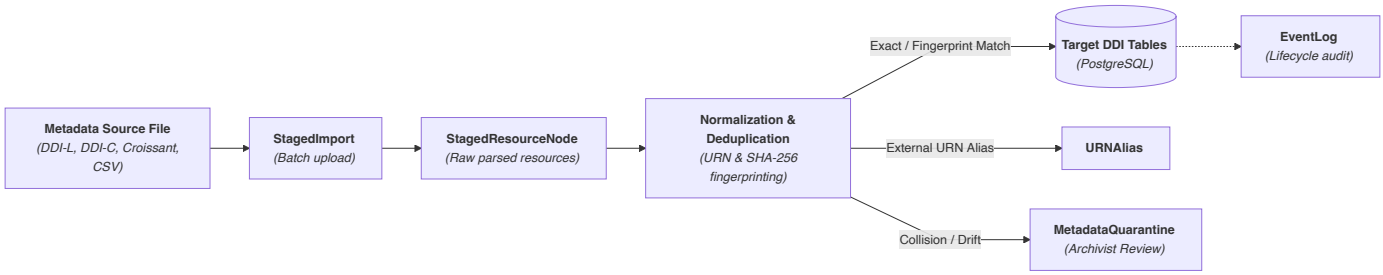
EventLog

Generic lifecycle mutation and audit log for any DDI-L resource.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing ID
`urn`	`CharField(512)`	indexed	Canonical or source URN of target resource
`timestamp`	`DateTimeField`	auto, indexed	Event occurrence timestamp
`event_type`	`CharField(128)`	indexed	Event type (`'created'`, `'normalized'`, `'quarantined'`)
`event_data`	`JSONField`	default `{}`	Arbitrary payload metadata as JSONB

8. Infrastructure, Staging & Ingestion Layer

The Ingestion layer provides a crash-proof, streaming staging pipeline for multi-standard metadata files (DDI-Lifecycle 3.3, DDI-Codebook 2.5, Croissant, CSV).



8.1 Table Definitions

UrnRegistry

Central master supertype entity table mapping canonical URNs and external URIs to their target resource types. Serves as the primary key foreign-key constraint target for all DDI entity tables (`REFERENCES request_ddi_urnregistry(urn) ON DELETE CASCADE`) and polymorphic reference tables (`URNAlias.canonical_urn`, `SemanticRelationship.subject_urn/object_urn`, `EventLog.urn`), guaranteeing global URN uniqueness and $O(1)$ crosswalk resolution for DDI and non-DDI identifiers.

Column	Type	Constraints	Notes
`urn`	`CharField(512)`	PK	Canonical URN or external identifier/URI (referenced by all entity tables)
`resource_type`	`CharField(64)`	indexed, required	Target entity/resource type (`"QuestionItem"`, `"Category"`, `"Concept"`, etc.)
`extended_attributes`	`JSONField`	default `[]`	Extensible attributes array of objects
`created_at`	`DateTimeField`	auto	Record creation timestamp
`updated_at`	`DateTimeField`	auto	Record update timestamp

URNAlias

Maps auto-generated/random URNs from external tools to canonical database entities.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing ID
`alias_urn`	`CharField(512)`	unique	External / random source URN
`canonical_urn`	`CharField(512)`	indexed	Canonical database URN
`entity_type`	`CharField(64)`	required	DDI entity type name
`hash_strategy`	`CharField(64)`	default `"v1_strict_sha256"`	Strategy used to establish link
`source_file`	`CharField(512)`	nullable	Originating import file
`created_at`	`DateTimeField`	auto	Record creation timestamp

MetadataQuarantine

Holds incoming metadata that requires archivist review due to URN collisions or drift.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing ID
`incoming_urn`	`CharField(512)`	required	Conflicting incoming URN
`existing_urn`	`CharField(512)`	nullable	Conflicting existing URN
`entity_type`	`CharField(64)`	required	Target DDI entity type
`incoming_content`	`JSONField`	required	Full serialized incoming payload
`existing_content_hash`	`CharField(64)`	nullable	Hash of existing record
`incoming_content_hash`	`CharField(64)`	required	Hash of incoming record
`conflict_type`	`CharField(32)`	required	`"staged_urn_drift"`, `"hash_mismatch"`
`resolution`	`CharField(32)`	nullable	`"approved"`, `"forked"`, `"rejected"`
`resolved_by`	`CharField(255)`	nullable	Reviewing archivist username
`resolved_at`	`DateTimeField`	nullable	Review timestamp
`source_file`	`CharField(512)`	nullable	Originating file
`import_task_id`	`CharField(255)`	nullable	Task queue reference
`created_at`	`DateTimeField`	auto	Record creation timestamp

StagedImport

Stores metadata and parameters for uploaded metadata file bundles and batch import jobs.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing import ID
`source_format`	`CharField(64)`	required	`"ddi-l:3.3:xml"`, `"ddi-l:4.0:json"`, `"ddi-c:2.5:xml"`, `"croissant"`, `"csv"`
`file_name`	`CharField(512)`	required	Original uploaded filename
`file_path`	`FileField`	nullable	File storage path
`import_options`	`JSONField`	default `{}`	Batch configuration metadata
`total_resources`	`IntegerField`	default 0	Total staged resource nodes
`processed_resources`	`IntegerField`	default 0	Processed resource count
`status`	`CharField(32)`	default `"staged"`	`"staged"`, `"normalized"`, `"quarantined"`, `"failed"`
`import_task_id`	`CharField(255)`	nullable	Background queue ID
`created_at`	`DateTimeField`	auto	Record creation timestamp
`processed_at`	`DateTimeField`	nullable	Completion timestamp

StagedResourceNode

Stores individual broken-down raw element resources extracted during Stage 1 parsing.

Column	Type	Constraints	Notes
`id`	`BigAutoField`	PK	Auto-incrementing node ID
`staged_import_id`	`BigInt`	FK → StagedImport	Parent staged import bundle
`resource_type`	`CharField(64)`	db_index	`"QuestionItem"`, `"CodeList"`, `"Category"`, `"StudyUnit"`, etc.
`raw_urn`	`CharField(512)`	db_index	Raw external URN from source file
`raw_value`	`JSONField`	required	Pre-normalized raw JSON payload
`canonical_urn`	`CharField(512)`	nullable, db_index	Resolved canonical database URN
`status`	`CharField(32)`	default `"staged"`	`"staged"`, `"normalized"`, `"quarantined"`, `"failed"`
`created_at`	`DateTimeField`	auto	Record creation timestamp

9. Index Strategy & Performance

Table	Index Column(s)	Type	Purpose
		PRIMARY	

All DDI entities	`urn`	KEY (B-tree)	Direct $O(1)$ / $O(\log N)$ identity lookups & FK referencing
`Concept`	`uri`	UNIQUE B-tree	Controlled vocabulary URI lookup
`Concept`	`vocabulary`	B-tree	Filter concepts by scheme
`Concept`	`notation`	B-tree	Notation/code lookup
`Code`	`(code_list_urn, code_value)`	UNIQUE composite	Code uniqueness within a code list
`QuestionVariable`	`(question_item_urn, represented_variable_urn)`	UNIQUE composite	Question to variable association uniqueness
`Datafile`	`urn`	PRIMARY KEY (B-tree)	Identity lookup & variable grouping
`Datafile`	`study_unit_urn`	B-tree	Study wave datafiles lookup
`DatafileVariable`	`(datafile_urn, instance_variable_urn)`	UNIQUE composite	Datafile to variable junction uniqueness
`DatafileVariable`	`(datafile_urn, order)`	Composite B-tree	Ordered column variable retrieval
`DatafileVariableFrequency`	`(datafile_urn, instance_variable_urn)`	Composite B-tree	Variable distribution retrieval per datafile
`DatafileVariableFrequency`	`(instance_variable_urn, code_value)`	Composite B-tree	Category code frequency lookup
`StudyInstrument`	`(study_unit_urn, instrument_urn)`	UNIQUE composite	Study to instrument mapping uniqueness
`EventLog`	`(urn, timestamp)`	Composite B-tree	Chronological audit log lookups by resource URN
`EventLog`	`event_type`	B-tree	Filter audit events by classification
`UrnRegistry`	`resource_type`	B-tree	Fast polymorphic resource type filtering
`URNAlias`	`alias_urn`	UNIQUE B-tree	Alias resolution during ingestion
`URNAlias`	`canonical_urn`	B-tree	Reverse alias lookup
`MetadataQuarantine`	`resolution`	Partial (where null)	Pending archivist review queue
`StagedResourceNode`	`raw_urn`, `canonical_urn`	B-tree	Fast node matching